

Session-based Music queue Reordering System Using Skip Behaviour as Implicit Feedback ~ Proposal

Data Science MSc

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Introduction

Over the past 15 years, music streaming services such as Spotify, Apple Music and YouTube have significantly altered the way our society consumes music. With over 100 million tracks now at the touch of our fingertips the challenge for these services has been how to enable the user to browse this plethora of options to find what most fits in with our personal tastes.

While these recommender systems have greatly advanced, they mostly focus on finding long-term patterns to suggest new songs that suit the user's current listening habits. One blind-spot that has arisen is modelling what the user wants to listen to in a particular session of listening, based on their mood and context.

Consider this scenario, which I am sure the reader would have also experienced. A user opens Spotify, unsure of exactly what song fits their mood, they begin shuffling through their liked songs. Skip after skip occurs (skipping songs that they may generally like, just not in this precise moment), occasionally they find a song they like and listen all the way through. This flow is then interrupted with another shuffled song that doesn't suit the context. These skip/full play signals are painting an accurate picture of what music the user currently wants but the current long-term approach is ill-equipped to capture such patterns.

This project will attempt to fill this gap, by constructing a recommender system that utilises the implicit feedback derived from the users skip behaviour to dynamically reorder the playlist queue in simulated real time, with the hopes of creating a more continuous listening experience. By treating skips as negative feedback and full plays as positive feedback the model will estimate the probability of each upcoming song being skipped and rank them with an aim of reducing skip-likelihood. The model will observe each skip/full-play signal as it would in a live listening session, estimating the skip-probabilities of each unplayed song after each listening event.

Literature Review

Music recommender systems have become an important component of modern streaming platforms. A seminal piece of research in this area was put forward by *Schedl et al. (2018)*. Their research compared two subjects: collaborative filtering, which predicts user preference on behavioural similarities between users and content-based filtering which recommends songs based on their acoustic features. Collaborative filtering has gone on to be one of the leading methods for recommending music, however it relies on long-term interaction history and cannot adapt to short-term mood-driven preferences within a listening session. Content-based filtering, on the other hand, recommends new songs on audio features alone but lacks the context to respond to what a user wants in each moment. The limitations addressed in both methods contribute to the session-based approach carried out in this project.

The use of implicit signals (such as plays, skips and completions) rather than explicit ratings as the basis for preference modelling in recommender systems was formalised by *Hu et al. (2008)*, who demonstrated that passive behavioural signals could be treated as noisy

indicators of preference with varying degrees of confidence. Whilst this established implicit feedback as an effective signal for recommendation, it assumed that preference is stationary, meaning that the signals a user generates over months of listening reliably predict what they want to hear in any given moment. As described in the introduction, this assumption breaks down in the context of music streaming, where a user's mood and context vary considerably from session to session.

The commercial significance of within-session skip signals was demonstrated by *Moor et al. (2023)*, whose large-scale study at Spotify showed that reordering a playlist queue based on sequential skip preferences measurably reduced skip rates across millions of users. However, the modelled skip probability is estimated from just historical behavioural patterns. Different songs are all treated as equal data points with no information about their acoustic properties. This results in the model identifying that users tend to skip at certain positions in a session. However, it can't deduce whether a skip was caused by a mismatch in audio features desired at that time or personal reasons unrelated to the audio content, such as a song associated with a bad memory or one that has been overplayed. Without audio features the model may struggle to generalise skip patterns to songs that are acoustically similar but not previously encountered.

The why of users skipping music and what features are most predictive of skipping behaviour was investigated by *Meggetto et al. (2023)*, who adapted Deep Reinforcement Learning for sequential music skip prediction using the Spotify Dataset. Using Shapley Additive Explanations (SHAP) and ablation analysis, they divided input features into three categories; behavioural, content and contextual. They found that behavioural features, specifically the reason a track started and pause behaviour between playbacks were the most significant predictors of skipping. Proposing that the most likely reason for a particular song being skipped is due to whether the user is in a restless/skipping mood, a state of skip-momentum. On the surface the research carried out by Meggetto et al. may seem to contradict the motivation behind this project, however their model was tested on each feature set in isolation so cannot comment on the combined effects of these features. Moreover, the binary skip-classification carried out in their work may be able to predict whether the next few songs will be skipped. This, however, does not propose any way to actively reduce the skip probabilities over the next few songs. This project addresses this by extending it to the environment of a ranking problem; dynamically reordering the queue according to the skip probabilities of the unplayed songs, using audio and contextual features to distinguish between songs which best match the users current mood, regardless of their skip-momentum state.

The most directly comparable work this author could find was the work of *Jin and Lee (2024)*, who proposed a late fusion neural network architecture for music skip prediction that combines audio features with contextual features. Evaluated on Spotify's Music Streaming Sessions Dataset (MSSD) *Brost et al (2019)*, their model achieved a Mean Average Accuracy (MAA) of 0.6713, which significantly outperformed their baseline models. Their work validates the significance of combining audio features with session context for skip-prediction, however I will go on to address a couple of limitations still present in their work.

When training at session-level, a context window is set at exactly 10 songs, meaning the model is trained on the first 10 songs and evaluated on the remainder. This makes it inflexible to varying session lengths, which can be anywhere from 10 songs - 100. Similarly to Meggetto et al., Jin and Lee also frame the task as solely a binary classification task, with no mention of ranking quality and their loss function investigation is limited to standard binary cross-entropy. Furthermore, their project utilises the MSSD which was taken down by Spotify in November 2024, making it no longer reproducible.

The framing of skip prediction specifically as a ranking problem was addressed by *Seshadri et al. (2024)*. They proposed a contrastive learning sub-task for sequential music recommendation that structures item embeddings so that skipped tracks are ranked lower across the session. Their experiments demonstrated improvements in item ranking and skip down-ranking across multiple datasets and across architectures including the Self-Attentive Session Recommendation Model (SASRec) *Kang & McAuley (2018)* and Bidirectional Encoder Representations from Transformers for sequential Recommendation (BERT4Rec,) confirming that transformer-based sequential models are well suited to the skip prediction task. Further, this shows that encoding skip signals directly into the ranking objective produces meaningfully better ordered queues than treating classification accuracy as a stand-in for ranking quality. However, like Moor et al., Seshadri et al. operate on behavioural aspects with no audio features, and their contrastive objective is not designed for factors such as position in session or skip intensity.

Finally, the question of which loss function should be applied to the model considering the noisy nature of implicit feedback is addressed by *Zhang et al. (2025)*. They identified structural weaknesses in the dominant loss functions of collaborative filtering and proposed the Distributional Robust Rényi Loss (DrRL). Their analysis found that Softmax Loss, whilst effective, is overly sensitive to false negative instances due to its use of KL-divergence in its underlying optimisation framework. Additionally, Cosine Contrastive Loss suffers from poor data utilisation because of its truncation mechanism filtering out most negative training samples. DrRL addresses both by replacing the divergence metric with the more flexible Rényi divergence, which allows finer control over how the model weights negative instances during training. This is particularly relevant to skip prediction, where both false positives and false negatives are common. A completed play may be a result of distraction rather than genuine preference (false-positive), whilst a skip may be caused by personal reasons unrelated to the song's audio properties (false-negative). DrRL's tolerance to such noise makes it a strong candidate for this setting. However, it was developed for general collaborative filtering and therefore is not suitable to be directly applied to a skip-based sequential recommendation system. Therefore, an adapted version of the DrRL will be applied and evaluated against other loss functions.

Existing research demonstrates that skip behaviour can be regarded as a valuable implicit signal, can be used to model short-term listening habits and can show that sequential architectures are a suitable for session-aware recommendations tasks. However previous works have largely approached skip prediction as a binary-classification problem rather than a dynamic ranking problem and behavioural-only approaches may not fully capture the whole

scope of aspects influencing a skip. Moreover, the noisy and context-dependent nature of implicit feedback has often been glossed over. Therefore, this project will investigate whether sequential models incorporating both behavioural and audio features can improve the accuracy of skip-likelihood predictions and therefore in-session queue reordering. Additionally, it will investigate whether a novel weighted loss-function can improve ranking quality under noisy conditions.

Methodology

Datasets

Two datasets will be featured within this project;

Firstly, Audio features for each track as follows: (key, mode, tempo, danceability, energy, loudness, speechiness, acousticness, instrumentality, liveness & valence). This will be sourced from a dataset of 56.3 million songs available through [huggingface.co Le Droff, G. \(2025\)](https://huggingface.co/LeDroff/G256), compiled from Spotify's now restricted audio features API.

The second is session-level behavioural data collected from Spotify's Extended Streaming History Export (ESHE) *Spotify A.B. (2026)*, available to all Spotify users. The metadata includes timestamps, milliseconds played, the reason a track began, the reason a track ended and whether the user was in shuffle mode. This data will be collected from a minimum of 30 volunteers from a varied age range and demographic. Based on preliminary exploratory analysis using this author's own export, this will sum to an estimated 80,000-100,000 usable sessions after filtering.

Volunteers will be recruited and informed clearly of what their data will be used for, how it will be stored, how long it will be stored for and how it will be anonymised. A clear indication of consent will be recorded. Upon reception, a random UUID will be assigned to each volunteer's data before being combined with the rest of the dataset. All raw data will be deleted after the submission and review of the project.

Sessions will be constructed by grouping consecutive listening events separated by gaps of no longer than 30 minutes. This length of time is consistent with the commonly accepted inactivity threshold in session-based recommendation research *Halfaker et al., (2015) [12]*. Hour of day and day features will be derived from the ESHE timestamps.

These sessions will then go through a process of filtering to remove any potential outliers, the following outlines what these filters consist of, in the presented order;

- **Any songs that were not shuffled;** indicating intentional song selection rather than mood driven response
- **Any session containing less than 10 songs;** this number was chosen as any less than this will not leave sufficient data to train on. A higher threshold would reduce the size of the dataset excessively.
- **Any session containing exactly 0 skips,** at least 1 skip is assumed to indicate attentive listening.

- **Any session containing only songs from the same artist**, indicating intentional artist selection and a lack of sufficient variance in audio features for meaningful prediction.

After filtering, sessions will be joined with the Le Drogoff dataset on Spotify's unique track identifier. Songs present in the ESHE but absent from the Le Drogoff dataset will be removed. Continuous features will be normalised to a mean of 0 and standard deviation of 1. Categorical features such as key will be one-hot encoded whilst mode will be treated as binary.

Model Architecture

This project will evaluate models across 2 tiers of complexity;

Baseline Models

The first tier comprises of two clearly interpretable baseline models, both consisting of the following inputs; feature vector of audio features & contextual features per song.

- **Logistic Regression:** A probabilistic classifier with feature coefficients that will illuminate any features promoting skipping across the population.
- **Random Forest:** Captures any non-linear relationships the logistic regression model may miss. This model involves a secondary output containing feature importances, which will give an indication as to how much each feature explains the variance in skip-probability.

Neural Networks

The second tier will consist of two neural networks, both consisting of a sequence of feature vectors across each session and will output a skip probability for each remaining song.

- **Recurrent Neural Network (RNN):** Introduction of a recurrent hidden state that is updated after every listening event so that the model can process each session as an ordered sequence and analyse any sequential patterns between consecutive songs.
- **Self-Attentive Sequential Recommendation model (SASRec):** This extends the sequential approach of the RNN by replacing the hidden state with a causal self-attention mechanism, this allows each position in the session to attend all prior listening events simultaneously. Training sessions are processed with a causal mask, that will ensure each position can only visit prior events, preventing any future data leakage. This architecture has been chosen for this research because of the work done by *Seshadri et al. (2024)*.

Queue Reordering

Each of the 4 models will output predicted skip-probabilities of every unplayed song remaining in the queue. These will be used to dynamically reorder the queue in ascending order of predicted skip-likelihood after each listening event. This will be the primary output

of the system. This distinguishes this project from the works of Meggetto et al. and Jin and Lee which don't address the consequential queue ordering problem.

Training

Prior to training, each individual's sessions will be split randomly upon training, validation and testing datasets according to a 70/15/15 ratio. This will reduce bias by ensuring that all 3 splits contain data from every volunteer and that the model won't be trained on a handful of individuals and therefore overfit according to their unique patterns. The sequential models will be trained on full sessions with SASRec's causal mask preventing future data leakage without requiring a manual session split. Hyperparameters will be tuned on the validation set with early stopping.

Loss Function Investigation

In addition to the 4 model architectures outlined above, this project will be assessing whether a custom loss function can help to improve the accuracy of skip prediction within the sequential models. These 3 loss functions are as follows:

- **Standard Binary Cross-Entropy (BCE) ~ Baseline:** This is the default loss function used in binary classifications tasks *Jin and Lee*. It treats all prediction errors equally. Zhang et al addresses the limitations of such a loss function.
- **Adapted Distribution Robust Rényi Loss (DrRL):** Proposed by Zhang et al., DrRL was designed to address the weaknesses experienced with loss functions when training on noisy implicit feedback. This function was developed for collaborative filtering so cannot be applied directly to session-based sequential recommendation so therefore will be adapted for this setting.
- **Custom Position-Weighted Time-Sensitive Loss (PWTS):** A novel loss function proposed by this project, designed to incorporate two aspects of skip data that BCE and DrRL do not account for;
 - o A position weight that penalises prediction errors on early songs more heavily, motivated by the works of Moor et al. who state that position in a session is a determinate predictor of skip behaviour.
 - o A time-sensitive weight derived from the milliseconds of the track played compared to the track duration, this will penalise skips that occur earlier in a song.

Evaluation

Skip-Probability Evaluation

Each session will be randomly split whilst preserving the chronological order. Minimum thresholds will be applied to the number of skips that are present in the context segment. The

model will be shown only the context segment of the session to simulate a model operating in real-time and then evaluated on the skip-probabilities for the remaining songs. Each model will be assessed using AUC-ROC as a classification metric measuring skip-probability accuracy, this is motivated by the presence of this metric in prior skip prediction work *Meggetto et al. [4], Jin & Lee.*

Queue Reordering Evaluation

Mean Reciprocal Rank (MRR) and Normalised Discounted Cumulative Gain (NDCG) will be used as a ranking-evaluation metrics, consistent with sequential music recommendation research. *Seshadri et al, Pereira et al. (2019)*

Pereira et al. reinforces the use of both classification and ranking metrics to evaluate music recommendation ranking problems in their work. Each model will utilise identical training, validation and evaluation splits with the exact same feature set, to ensure a fair comparison.

Project Timeline

The project is structured across six phases spanning May to September 2026, with a submission deadline of September 11th and a three-week buffer period beginning August 24th to absorb any slippage.

Phase 1: Data Collection and Preprocessing (May 6th to June 10th) Volunteer outreach begins in early May to account for Spotify's data export delays, which can take up to 30 days. The preprocessing and filtering python scripts can be written and tested using the author's own export data before all volunteer data has arrived, ensuring no unnecessary idle time during the collection window.

Phase 2: Baseline Model Development (June 10th to June 17th) Logistic Regression and Random Forest are implemented first, providing interpretable benchmarks before the more complex sequential models are introduced. Their feature coefficient and importance outputs also inform feature understanding ahead of the neural network phase.

Phase 3: Neural Network Development (June 18th to July 9th) The RNN is implemented first, with SASRec building directly on its infrastructure. This ordering is deliberate, as debugging at the RNN stage reduces friction when implementing the more complex causal self-attention mechanism. On days where models are actively training, time can be used on parallel tasks such as report writing or hyperparameter planning on previous models, hence the overlapping. This means training time does not represent idle time in the project schedule.

Phase 4: Loss Function Investigation (July 10th to July 23rd) The loss function investigation is scheduled immediately after the core models are functional, as it depends on the RNN and SASRec training loops already being in place. Similarly, training runs during this phase free up time for drafting the results and evaluation chapters concurrently.

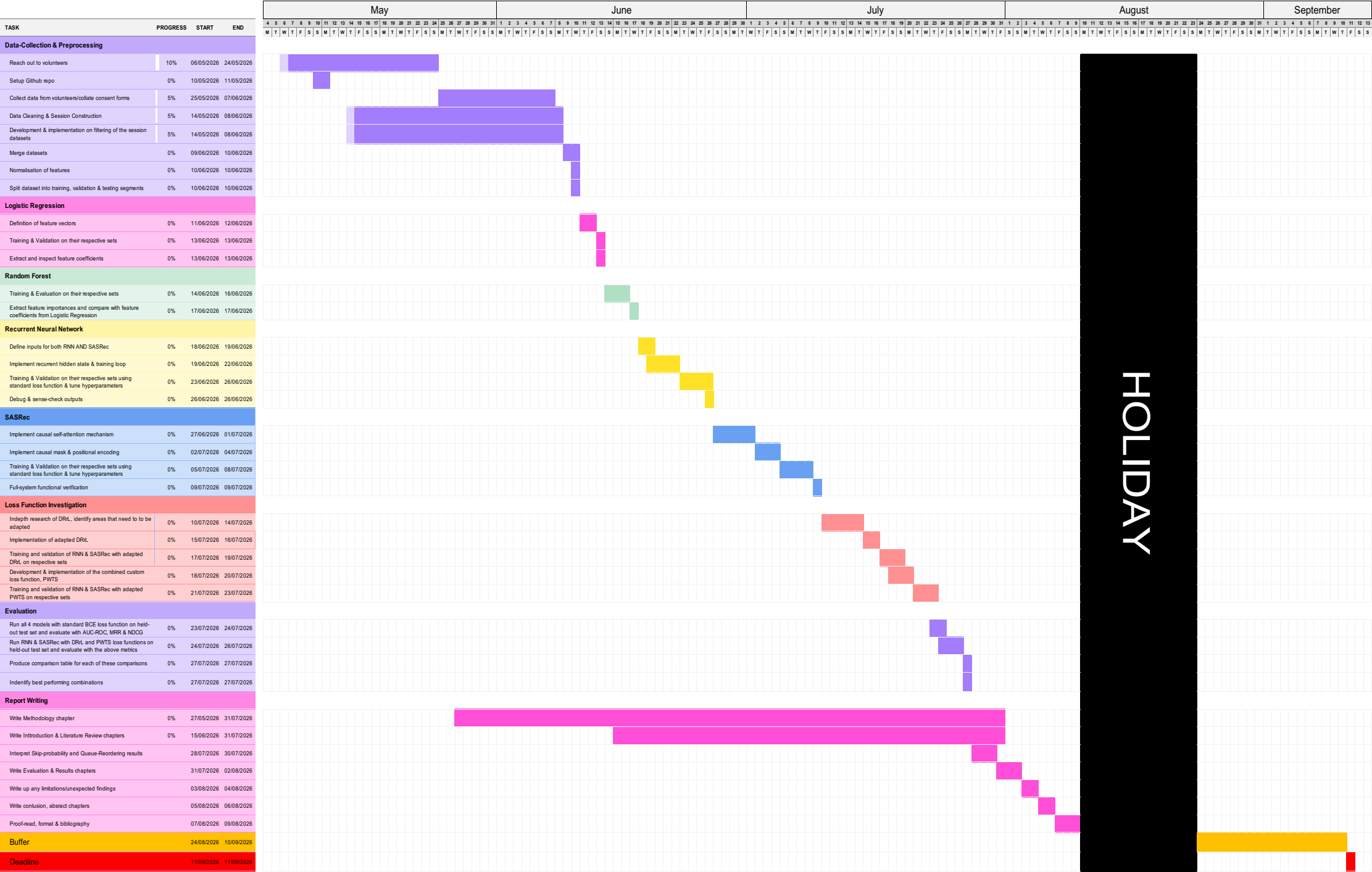
Phase 5: Evaluation (July 23rd to July 27th) Formal evaluation on the held-out test set takes place in late July, keeping test data strictly unseen until this point. All four models are first evaluated with standard BCE loss using AUC-ROC, MRR and NDCG, before the RNN and

SASRec are re-evaluated under the DrRL and PWTS loss functions. A comparison table is then produced and the best performing combinations identified.

Phase 6: Report Writing (Throughout) Report writing is distributed across the timeline rather than concentrated at the end. The methodology chapter is drafted from late May and the literature review from mid-June, reflecting content that is largely settled early in the project. Final chapters, proofreading, and formatting are completed in early August ahead of the buffer period.

Please see the next page for the Gantt chart outlining the above timeline.

Gantt Chart



Bibliography

- Halfaker, A., Keyes, O., Kluver, D., Thebault-Spieker, J., Nguyen, T., Shores, K., Uduwage, A. and Warncke-Wang, M. (2015) 'User session identification based on strong regularities in inter-activity time', in *Proceedings of the 24th International Conference on World Wide Web (WWW 2015)*. New York: ACM, pp. 410–418.
- Hu, Y., Koren, Y. and Volinsky, C. (2008) 'Collaborative filtering for implicit feedback datasets', in *Proceedings of the 8th IEEE International Conference on Data Mining (ICDM 2008)*. Piscataway: IEEE, pp. 263–272.
- Jin, S. and Lee, J. (2024) 'Effective music skip prediction based on late fusion architecture for user-interaction noise', *Expert Systems with Applications*, 238, p. 122098.
- Kang, W. and McAuley, J. (2018) 'Self-attentive sequential recommendation', in *Proceedings of the IEEE International Conference on Data Mining (ICDM 2018)*. Piscataway: IEEE, pp. 197–206.
- Le Droffoff, G. (2025) *Spotify huge track analysis dataset* [Dataset]. Hugging Face. Available at: <https://huggingface.co/datasets/GildasLeDroffoff/spotify-huge-track-analysis-dataset> (Accessed: 20 April 2026).
- Meggetto, F., Revie, C., Levine, J. and Moshfeghi, Y. (2023) 'Why people skip music? On predicting music skips using deep reinforcement learning', in *Proceedings of the ACM SIGIR Conference on Human Information Interaction and Retrieval (CHIIR 2023)*. New York: ACM, pp. 95–106.
- Moor, D., Yuan, Y., Mehrotra, R., Dai, Z. and Lalmas, M. (2023) 'Exploiting sequential music preferences via optimisation-based sequencing', in *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM 2023)*. New York: ACM.
- Pereira, B., Ueda, A., Penha, G., Santos, R. and Ziviani, N. (2019) 'Online learning to rank for sequential music recommendation', in *Proceedings of the 13th ACM Conference on Recommender Systems (RecSys 2019)*. New York: ACM, pp. 237–245.
- Schedl, M., Zamani, H., Chen, C., Deldjoo, Y. and Elahi, M. (2018) 'Current challenges and visions in music recommender systems research', *International Journal of Multimedia Information Retrieval*, 7(2), pp. 95–116.
- Seshadri, P., Shashaani, S. and Knees, P. (2024) 'Enhancing sequential music recommendation with negative feedback-informed contrastive learning', in *Proceedings of the 18th ACM Conference on Recommender Systems (RecSys 2024)*. New York: ACM.
- Spotify AB (2026) *Extended streaming history export* [Personal data file]. Available at: <https://www.spotify.com/account/privacy/> (Accessed: 27 April 2026).
- Brost, B., Mehrotra, R. and Jehan, T. (2019) 'The music streaming sessions dataset', in *Proceedings of the World Wide Web Conference (WWW 2019)*. New York: ACM, pp. 1179–1188. Available at: <https://arxiv.org/abs/1901.09851> (Accessed: 5 May 2026).
- Zhang, S., Chen, J., Li, C., Zhou, S., Shi, Q., Feng, Y., Chen, C. and Wang, C. (2025) 'Advancing loss functions in recommender systems: a comparative study with a Rényi divergence-based solution', in *Proceedings of the 39th AAAI Conference on Artificial Intelligence*. Menlo Park: AAAI Press, vol. 39, no. 12, pp. 13286–13294.